Course Code : ECT 357 **KQLR/RS – 24 / 1173**

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER NETWORKS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.

(2) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Enlist various layers of OSI Reference Model. What are the responsibilities ? Also the concerns of the data link layer in the model. 5(CO1)

   (b) Can you make a distinction between circuit switching and packet switching used in Computer Communications ? 5(CO1)

2. (a) What is purpose of transmission media in communication networks ? Briefly explain various Guided transmission media. 5(CO3)

   (b) Generate a CRC from the dataword 1010. Use 1001 as generator vector. Explain the process of error detection and correction of CRC with a suitable example. 5(CO3)

3. (a) Explain Go-Back-N ARQ protocol. Justify that Go-Back-N ARQ protocol is efficient than Stop-and-Wait ARQ protocol with reference to bandwidth utilization. 5(CO4)

   (b) Name the ATM layers and their functions. Explain, how is an ATM virtual connection identified. 5(CO4)

4. (a) Determine the shortest path from node **A** to all the nodes using Bellman Ford Algorithm for the network shown in **Fig. Q. 4 (a)**. Assume node **A** is a source Node.

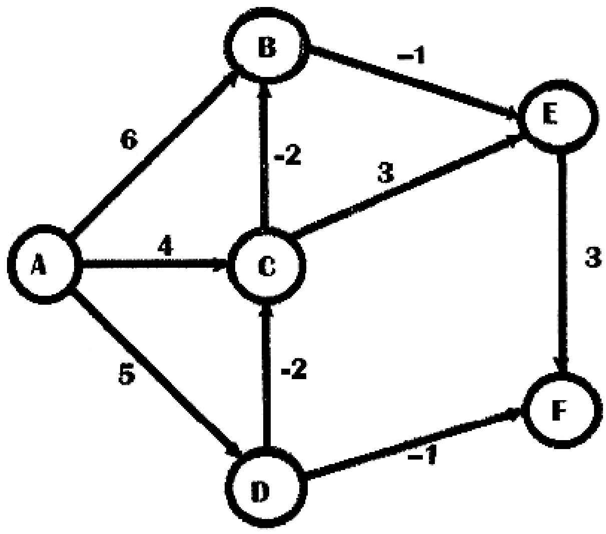

Fig. Q. 4 (a) 5(CO3)

(b) Design a Network for an organization which is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.

(i) Find the subnet mask.

(ii) Find the number of addresses in each subnet.

(iii) Find the first and last addresses in subnet 1.

(iv) Find the first and last addresses in subnet 32. 5(CO5)

5. (a) What is traffic shaping ? Name various methods to shape traffic. Explain any one method of traffic shaping. 5(CO4)

(b) Write short notes on :—

(i) Domain Name System.

(ii) Digital Signature. 5(CO2)

6. (a) Encrypt the message "THIS IS AN EXERCISE" using a shift cipher with a key of 20. Ignore the space between words. Decrypt the message to get the original plaintext. 5(CO2)

(b) For a RSA algorithm with p = 3 and q = 5, evaluate public key and private key. Also obtain the ciphertext for the plaintext **'LAKE'**. 5(CO2)